

BERGAMO ORIO AL SERIO, ItalyWMO: **160760**Lat: **45.669N**Lon: **9.700E**Elev: **237**StdP: **98.51**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.5	-12.8	1.3	0.6	-10.1	1.6	1.1	8.0	11.6	6.4	9.6	1.8	10	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	32.9	22.9	31.2	22.0	30.1	21.5	23.9	30.8	23.0	29.9	22.1	28.5	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.8	28.7	20.6	15.7	27.4	19.8	15.0	26.5	72.8	31.1	69.1	29.9	65.6	28.6	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.6	5.4	4.5	-6.2	34.5	2.1	1.5	-7.7	35.5	-8.9	36.4	-10.1	37.2	-11.7	38.3
			-7.5	25.0	2.0	1.2	-8.9	25.9	-10.1	26.6	-11.2	27.3	-12.6	28.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	3.9	5.2	9.5	13.3	17.8	22.1	24.2	23.8	19.3	14.5	8.9	4.5
	DBStd	7.83	2.74	3.01	3.32	3.18	3.23	3.30	2.56	2.79	2.82	2.99	3.20	2.96
	HDD10.0	614	189	135	48	8	0	0	0	0	0	3	58	172
	HDD18.3	2155	447	367	274	153	48	7	0	1	21	122	284	429
	CDD10.0	2061	1	1	32	106	243	362	441	428	279	143	23	1
	CDD18.3	562	0	0	0	2	33	118	183	171	51	4	0	0
	CDH23.3	4764	0	0	2	14	246	1033	1704	1478	275	12	0	0
	CDH26.7	1581	0	0	0	1	49	340	617	531	42	1	0	0
Wind	WSAvg	2.0	1.6	1.9	2.3	2.3	2.3	2.4	2.3	2.2	2.0	1.7	1.6	1.5
Precipitation	PrecAvg	958	51	49	51	89	106	92	68	89	103	108	111	57
	PrecMax	1710	250	186	167	185	220	284	199	198	369	262	349	159
	PrecMin	603	1	0	3	16	22	10	9	5	18	17	6	1
	PrecStd	273	52	51	38	41	56	62	50	52	81	63	90	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	17.0	22.1	25.1	30.2	33.6	34.2	35.0	29.5	24.8	18.2	13.7
		MCWB	8.0	10.0	12.8	16.1	20.2	23.6	22.9	23.9	20.6	18.0	13.3	8.8
	2%	DB	11.2	13.8	19.1	22.9	27.9	31.9	32.5	32.9	28.0	22.5	16.1	11.8
		MCWB	6.9	8.5	11.4	14.8	18.8	22.1	22.9	22.8	19.7	16.6	12.1	8.0
	5%	DB	9.8	12.0	17.2	21.2	26.0	30.1	31.1	31.1	26.3	21.0	14.7	10.0
		MCWB	6.2	7.1	10.8	13.9	17.5	21.4	22.1	21.9	18.8	16.0	11.8	7.4
	10%	DB	8.1	10.1	15.4	19.4	24.1	28.8	30.0	29.8	24.9	19.2	13.2	8.8
		MCWB	5.7	6.3	9.9	13.0	16.8	20.4	21.5	21.4	17.8	15.0	11.0	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	10.2	13.5	16.7	21.1	24.9	25.1	25.2	21.7	18.6	14.3	10.9
		MCDB	11.8	14.9	19.5	23.5	28.6	32.4	31.4	33.3	27.8	22.6	16.7	12.2
	2%	WB	8.1	9.0	12.2	15.4	19.4	23.0	23.8	23.9	20.6	17.6	13.2	9.2
		MCDB	9.6	12.4	17.5	21.5	26.3	30.2	30.4	31.0	26.0	21.0	15.0	10.3
	5%	WB	7.0	7.9	11.3	14.5	18.2	21.9	23.0	22.9	19.7	16.8	12.3	7.7
		MCDB	8.3	10.3	15.8	20.1	24.4	28.6	29.9	29.8	24.7	19.8	14.1	9.0
	10%	WB	5.9	7.0	10.4	13.5	17.2	21.0	22.2	21.8	18.7	15.8	11.5	6.8
		MCDB	7.4	9.0	14.3	18.4	23.1	27.7	28.6	28.0	23.5	18.7	13.2	8.1
Mean Daily Temperature Range		MDBR	7.0	7.8	9.0	9.0	9.5	9.8	10.0	9.7	8.8	7.3	6.3	6.7
	5% DB	MCDBR	9.5	11.0	12.0	11.6	11.7	11.1	10.7	10.8	10.3	9.4	8.0	8.7
		MCWBR	6.2	7.0	6.8	5.9	5.4	5.0	4.8	4.7	4.4	4.8	5.0	5.8
	5% WB	MCDBR	7.3	9.0	10.3	10.3	10.6	10.6	10.0	10.3	9.1	7.8	5.9	6.4
		MCWBR	5.3	6.1	6.1	5.5	5.2	5.0	5.0	5.1	4.3	4.3	4.1	4.8
Clear-Sky Solar Irradiance	taub	0.329	0.361	0.409	0.452	0.452	0.475	0.456	0.445	0.438	0.419	0.362	0.325	
	taud	2.383	2.286	2.181	2.097	2.151	2.130	2.191	2.231	2.218	2.232	2.357	2.420	
	Ebn at Noon	765	802	810	807	821	801	812	805	772	723	712	729	
	Edh at Noon	75	99	127	150	147	151	140	130	121	104	77	66	
All-Sky Solar Radiation	RadAvg	1.47	2.29	3.61	4.50	5.44	6.27	6.38	5.42	4.08	2.46	1.46	1.23	
	RadStd	0.21	0.32	0.47	0.54	0.47	0.39	0.41	0.36	0.30	0.36	0.25	0.17	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	-0.67	N/A	N/A	N/A
Regional (79 neighbors)		+0.35	N/A	N/A	+0.67	+0.39	+0.30	N/A	N/A	+61